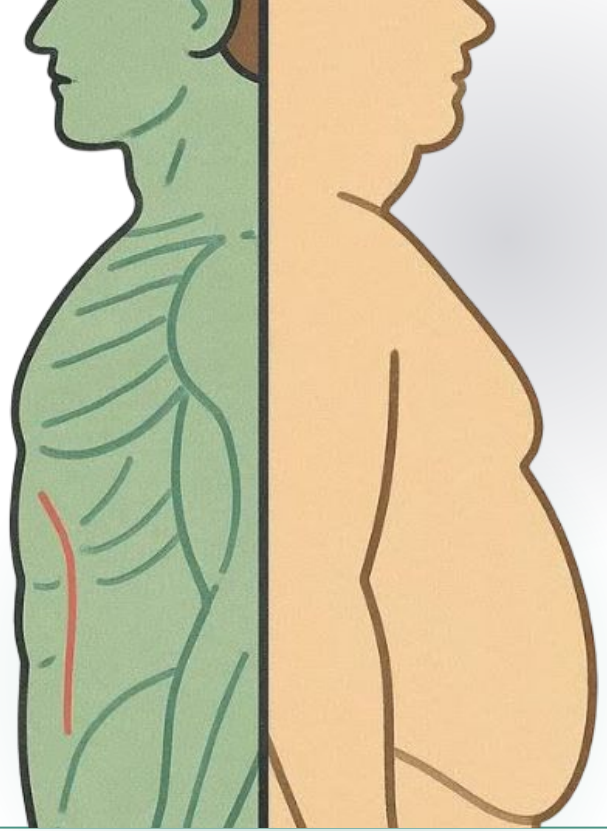
Predicting Body Fat %



Group 2: Clea Yenifer, Sutolimin Widjaja, Vincent Lavoie, Yuchi Yu

OUR OBJECTIVE:

Calculate Body Fat % using simple body measurements

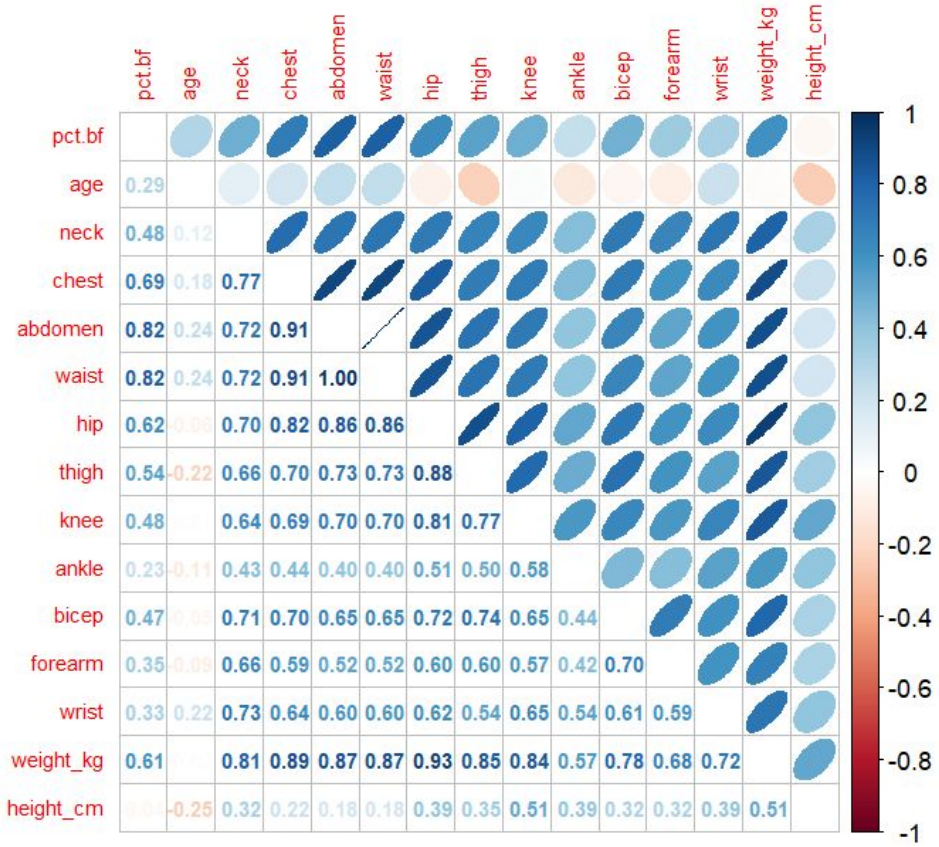
A simple, practical model that anyone can use - without sacrificing predictive power.

Dataset

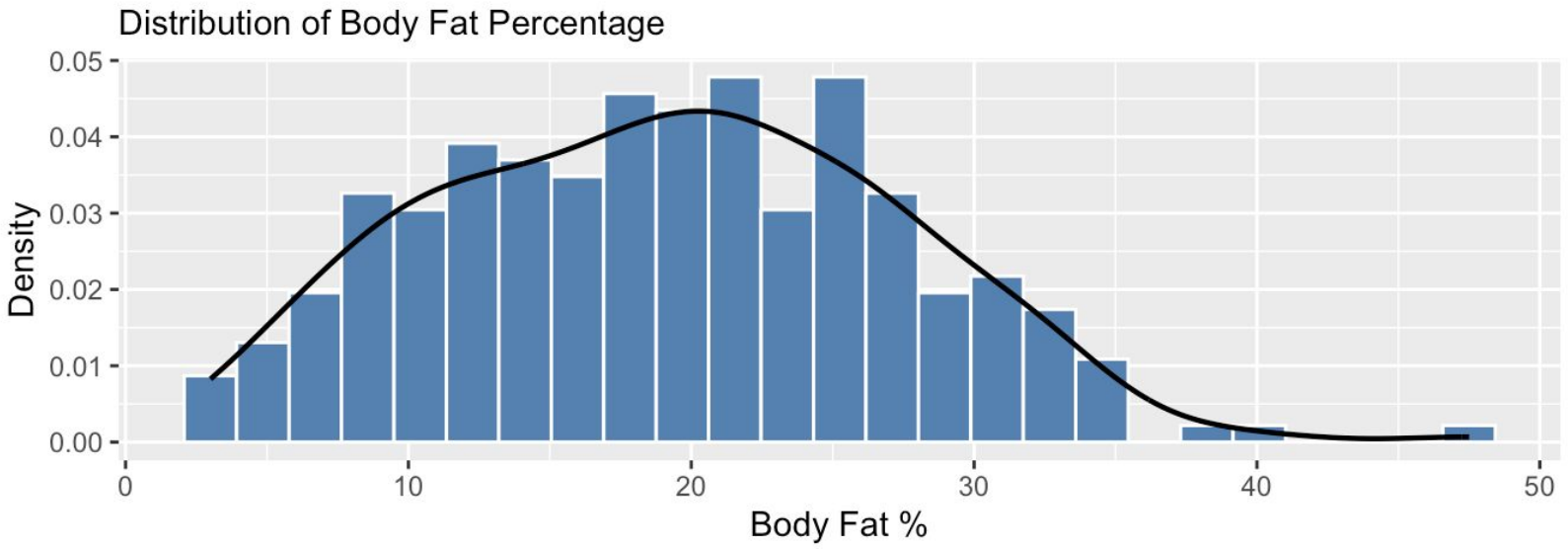
Density	Pct.BF	Age	Weight	Height	Neck	Chest	Abdomen
1.0708	12.3	23	154.25	67.75	36.2	93.1	85.2
1.0853	6.1	22	173.25	72.25	38.5	93.6	83
1.0414	25.3	22	154	66.25	34	95.8	87.9

Waist	Hip	Thigh	Knee	Ankle	Bicep	Forearm	Wrist
33.5433	94.5	59	37.3	21.9	32	27.4	17.1
32.6772	98.7	58.7	37.3	23.4	30.5	28.9	18.2
34.6063	99.2	59.6	38.9	24	28.8	25.2	16.6

EDA

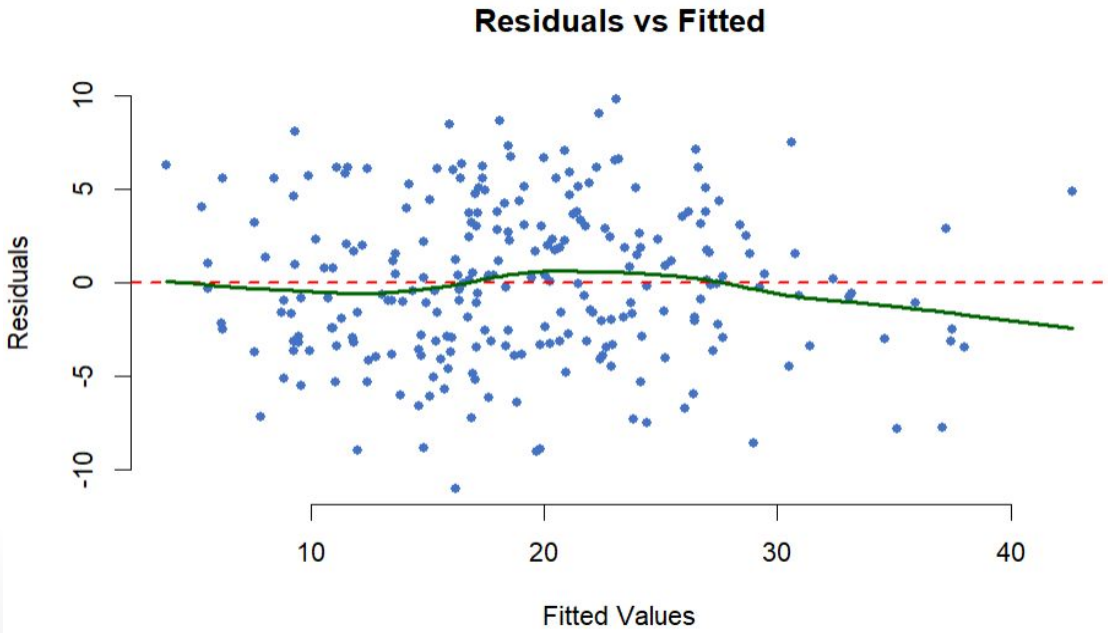

pct.bf
age
neck
chest
abdomen
waist
hip
thigh
knee
ankle
bicep
forearm
wrist
weight_kg
height_cm
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

EDA

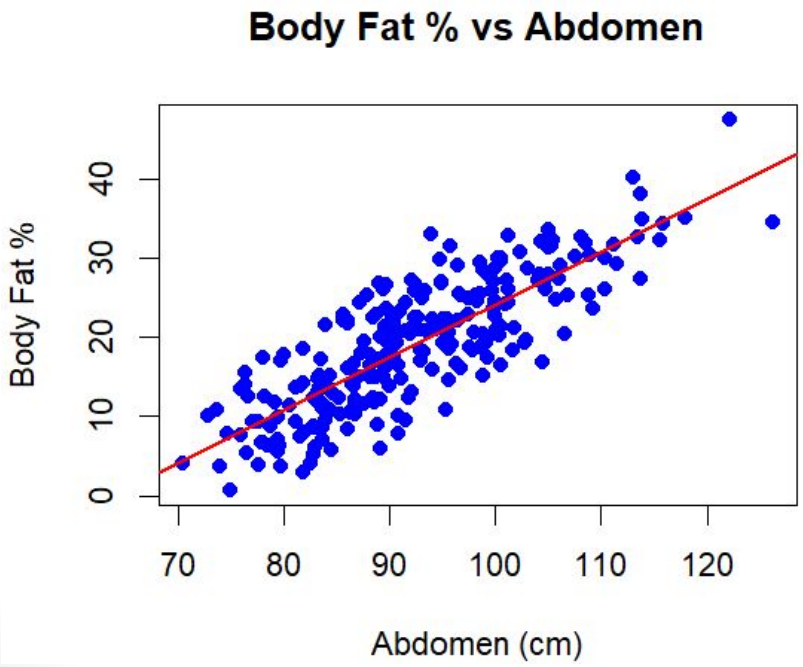

Distribution of Body Fat Percentage
Density
0.05
0.04
0.03
0.02
0.01
0.00
0
10
20
30
40
50
Body Fat %

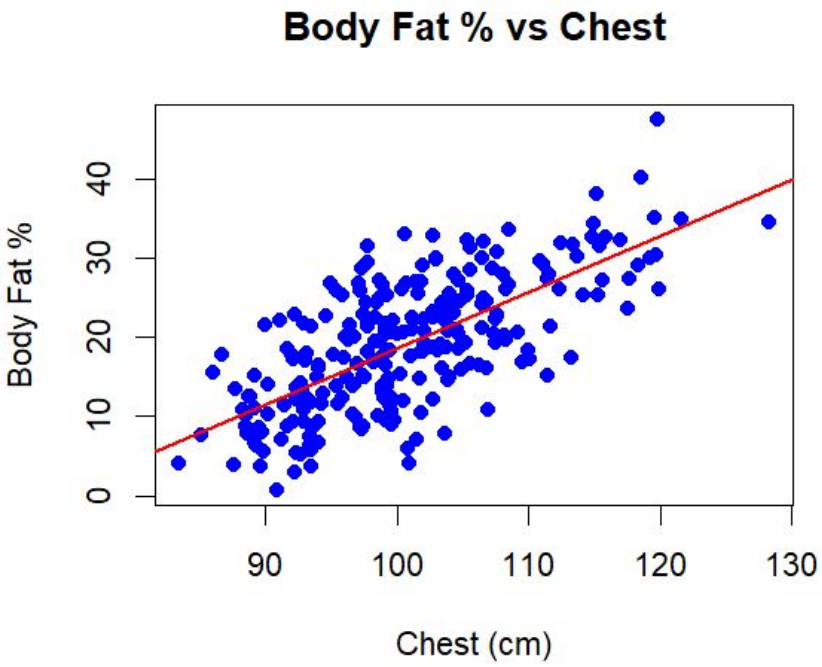
Model Experimentation

Body Fat % ~ age + neck + chest + abdomen + hip + thigh + knee + ankle + bicep + forearm + wrist + height_cm + weight_kg



Model Experimentation





Model Experimentation

VIF (Variance Inflation Factor)

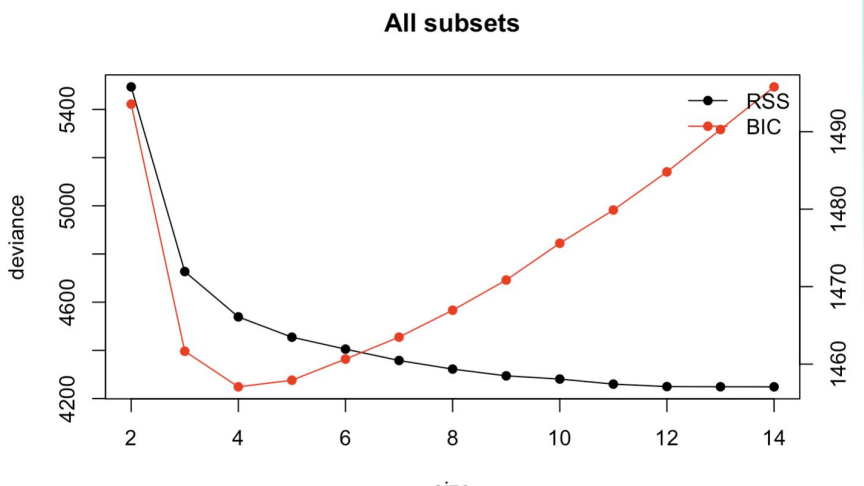
age	neck	chest	abdomen	hip	thigh	knee	ankle	bicep
2.28	3.93	10.63	11.73	11.9	7.08	4.36	1.79	3.39

bicep	forearm	wrist	weight_kg	height_cm
3.39	2.39	3.22	44.98	3.46

Body Fat % ~ age + abdomen + wrist + height_cm

Model Comparisons

1) Exhaustive Search



	age	neck	chest	abdomen	hip	thigh	knee	ankle	bicep	forearm	wrist	weight_kg	height_cm
1 (1)	" "	" "	" "	"*"	" "	" "	" "	" "	" "	" "	" "	" "	" "
2 (1)	" "	" "	" "	"*"	" "	" "	" "	" "	" "	" "	" "	"*"	" "
3 (1)	" "	" "	" "	"*"	" "	" "	" "	" "	" "	" "	"*"	"*"	" "
4 (1)	"*"	" "	" "	"*"	" "	" "	" "	" "	" "	" "	"*"	" "	"*"
5 (1)	"*"	" "	"*"	"*"	" "	" "	" "	" "	" "	" "	"*"	" "	"*"
6 (1)	"*"	" "	"*"	"*"	" "	" "	" "	" "	"*"	" "	"*"	" "	"*"
7 (1)	"*"	"*"	"*"	"*"	" "	" "	" "	" "	"*"	" "	"*"	" "	"*"
8 (1)	"*"	"*"	"*"	"*"	" "	" "	" "	" "	"*"	"*"	"*"	" "	"*"
9 (1)	"*"	"*"	"*"	"*"	"*"	" "	" "	" "	"*"	"*"	"*"	" "	"*"
10 (1)	"*"	"*"	"*"	"*"	"*"	"*"	" "	" "	"*"	"*"	"*"	" "	"*"
11 (1)	"*"	"*"	"*"	"*"	"*"	"*"	" "	"*"	"*"	"*"	"*"	" "	"*"
12 (1)	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	" "	"*"
13 (1)	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"

Model Comparisons

1) Exhaustive Search

```
Call:
lm(formula = pct.bf ~ age + abdomen + wrist + height_cm, data =
bodyfat)

Residuals:
     Min       1Q   Median       3Q      Max
-10.4887  -3.0515  -0.4114   3.3492   9.3245

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.69050    8.10188   0.456  0.64915
age          0.05629    0.02380   2.365  0.01882 *
abdomen      0.76726    0.03372  22.751  < 2e-16 ***
wrist       -1.92905    0.40938  -4.712 4.12e-06 ***
height_cm   -0.12781    0.04781  -2.673  0.00802 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.273 on 244 degrees of freedom
Multiple R-squared:  0.7343,    Adjusted R-squared:  0.7299
F-statistic: 168.6 on 4 and 244 DF,  p-value: < 2.2e-16
```

Body fat =
3.69+ 0.056age+0.76abdomen-1.929wrist-0.128height

P-value <0.05 => βi≠0

Model Comparisons

```r
n <- nrow(bodyfat)
M1 = lm(pct.bf ~ . -waist, data = bodyfat)

step.back.bic = step(M1,
                     direction = "backward",
                     k = log(n),
                     trace = FALSE)

#summary(step.back.bic)
round(summary(step.back.bic)$coefficients, 3)
```

2) Backward variable selection

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.690	8.102	0.456	0.649
age	0.056	0.024	2.365	0.019
abdomen	0.767	0.034	22.751	0.000
wrist	-1.929	0.409	-4.712	0.000
height_cm	-0.128	0.048	-2.673	0.008

P-value < 0.05

BIC vs. AIC: Using log(n) makes the model "stricter" than the standard AIC. It prefers a simpler model with fewer variables to avoid overfitting.

$$AIC = nlog\left(\frac{\text{Residual sum of squares}}{\text{n}}\right) + 2p$$

$$BIC = nlog\left(\frac{\text{Residual sum of squares}}{\text{n}}\right) + \log(\text{n})\text{p}$$

Model Comparisons

```r
M0 = lm(pct.bf ~ 1, data = bodyfat)

step.fwd.bic = step(M0,
                    scope = list(lower = M0, upper = M1),
                    direction = "forward",
                    k = log(n),
                    trace = FALSE)

#summary(step.fwd.bic)
round(summary(step.fwd.bic)$coefficients, 3)
```

3) Forward variable selection

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-27.259	6.743	-4.043	0.000
abdomen	0.957	0.055	17.334	0.000
weight_kg	-0.208	0.053	-3.897	0.000
wrist	-1.388	0.435	-3.186	0.002

P-value < 0.05

Exhaustive research is better than Backward/forward variable selection

Cross Validation

Model 1: Body Fat % ~ `age + abdomen + wrist + height_cm`

Model 2: Body Fat % ~ `abdomen + weight_kg + wrist`

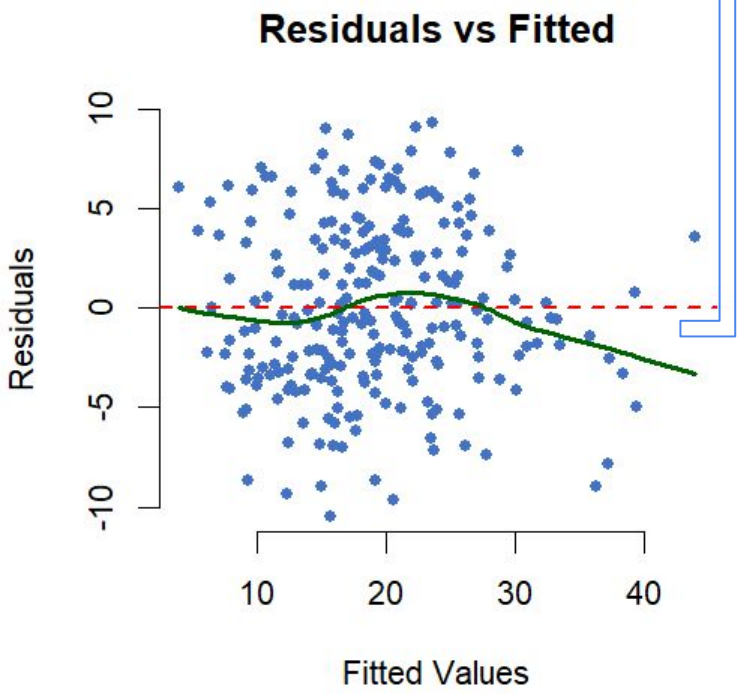
Model	RMSE	MAE	Rsquared
Model 1	4.292	3.559	0.724
Model 2	4.323	3.593	0.745

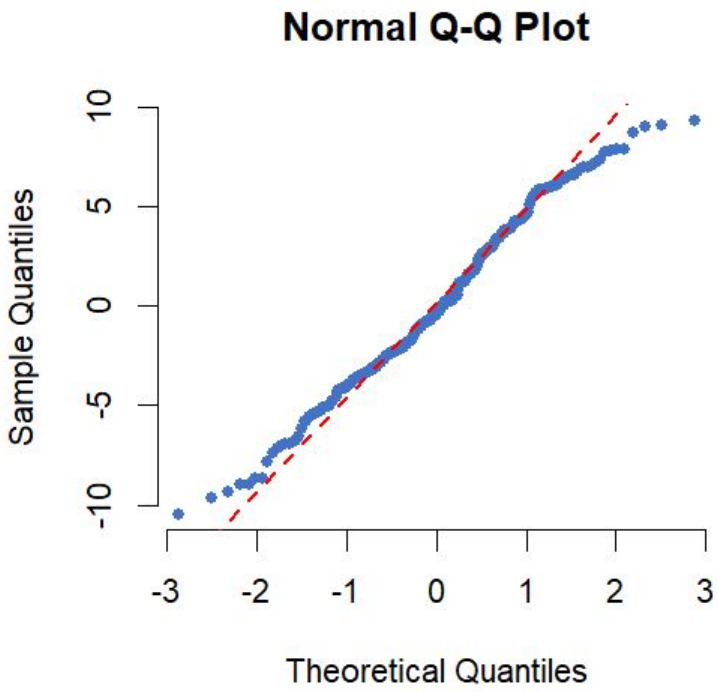
We choose Model 1 given the smaller errors, despite the lower R^2

Assumptions Checking

Linearity, Normality

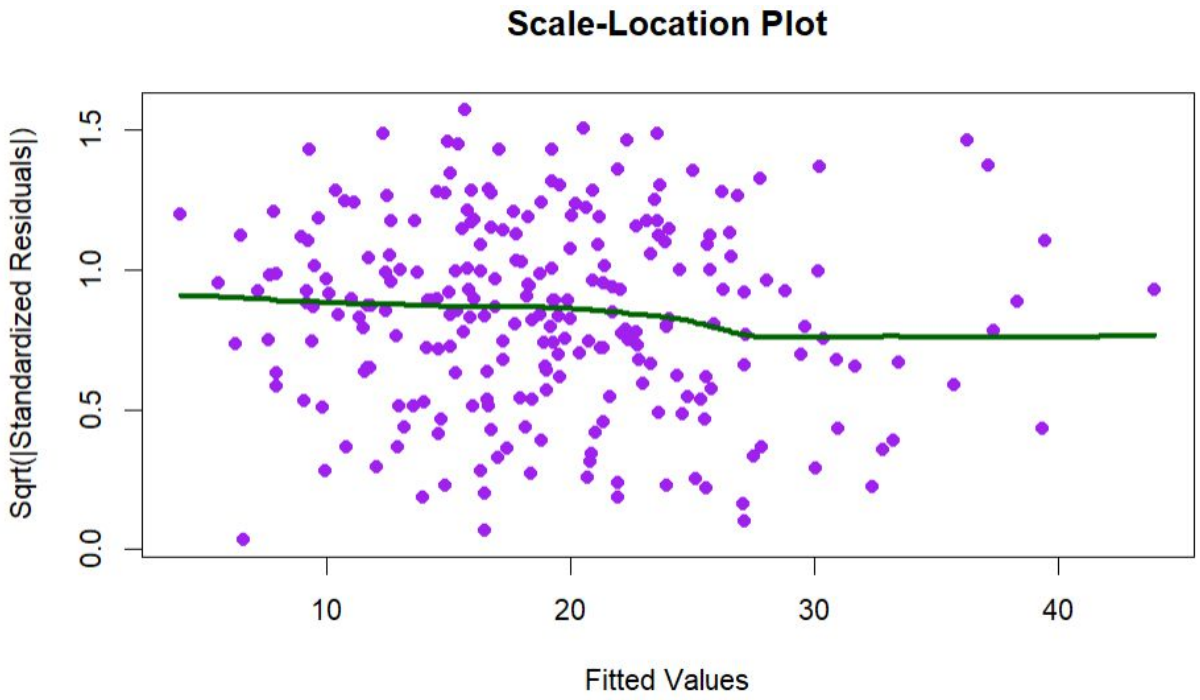
Overestimating the body fat percentage





Assumptions Checking

Independence, Homoskedasticity



Final Model

Body Fat % ~ age + abdomen + wrist + height_cm

```
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     Min       1Q   Median       3Q      Max 
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```

Limitations

01 Model predicts only male adults

02 Relatively small sample size

03 Measurement errors in the outcome

04 Tape measurement errors

Thanks!

Any Questions ?